

*Answer key***Problem 1.**

Convert each decimal number to hexadecimal:

$$26 = 1 \times 16 + 10 = 1A_{16}$$

$$60 = 3 \times 16 + 12 = 3C_{16}$$

$$175 = 10 \times 16 + 15 = AF_{16}$$

$$255 = 15 \times 16 + 15 = FF_{16}$$

Problem 2.

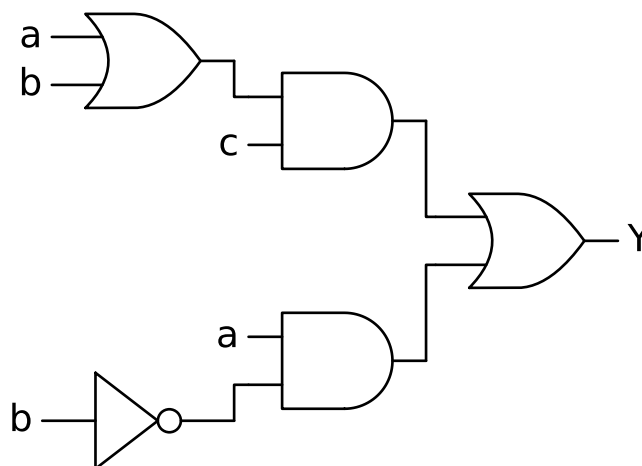
Convert each hexadecimal number to decimal:

$$1A_{16} = 1 \times 16 + 10 = 26$$

$$2F_{16} = 2 \times 16 + 15 = 47$$

$$B4_{16} = 11 \times 16 + 4 = 180$$

$$FF_{16} = 15 \times 16 + 15 = 255$$

Problem 3.

Expression: $(A + B) \cdot C + A \cdot \overline{B}$

Simplifies to: $B \cdot C + A \cdot \overline{B}$

Truth table:

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

Problem 4.

Convert each binary number to hexadecimal:

$$11010_2 = 00011010 = 1A_{16}$$

$$101111_2 = 00101111 = 2F_{16}$$

$$10110100_2 = 10110100 = B4_{16}$$

$$1111_2 = 1111 = F_{16}$$

Problem 5.

Convert each hexadecimal number to binary:

$$3C_{16} = 3 \rightarrow 0011, C \rightarrow 1100 = 111100_2$$

$$A7_{16} = A \rightarrow 1010, 7 \rightarrow 0111 = 10100111_2$$

$$1E_{16} = 1 \rightarrow 0001, E \rightarrow 1110 = 11110_2$$

$$FF_{16} = F \rightarrow 1111, F \rightarrow 1111 = 11111111_2$$

Problem 6.

Every device on a network has an *IP address*: four numbers separated by dots, each stored in one byte (so 0 to 255). Convert each part of these addresses to 8-bit binary.

192.168.1.1 = 11000000.10101000.00000001.00000001

10.0.0.255 = 00001010.00000000.00000000.11111111

8.8.8.8 = 00001000.00001000.00001000.00001000

Which one of these could *not* be a real IP address, and why?

192.168.0.1 10.0.256.4 172.16.0.255

Answer: 10.0.256.4 (256 is more than 255 — too big for one byte)

Fun fact: 8.8.8.8 is one of Google's real public addresses.

Problem 7.

A black-and-white picture is a grid of *pixels*: each pixel is one bit — 1 means a filled (black) square, 0 means blank. Each row of 8 pixels is one byte, given below in hexadecimal. Convert each byte to 8-bit binary, then shade a square wherever there is a 1.

66₁₆ = 01100110

FF₁₆ = 11111111

FF₁₆ = 11111111

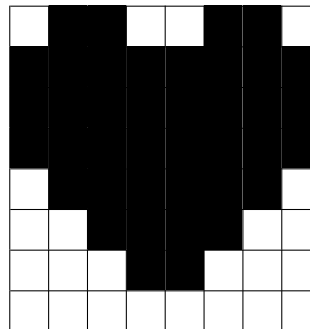
FF₁₆ = 11111111

7E₁₆ = 01111110

3C₁₆ = 00111100

18₁₆ = 00011000

00₁₆ = 00000000

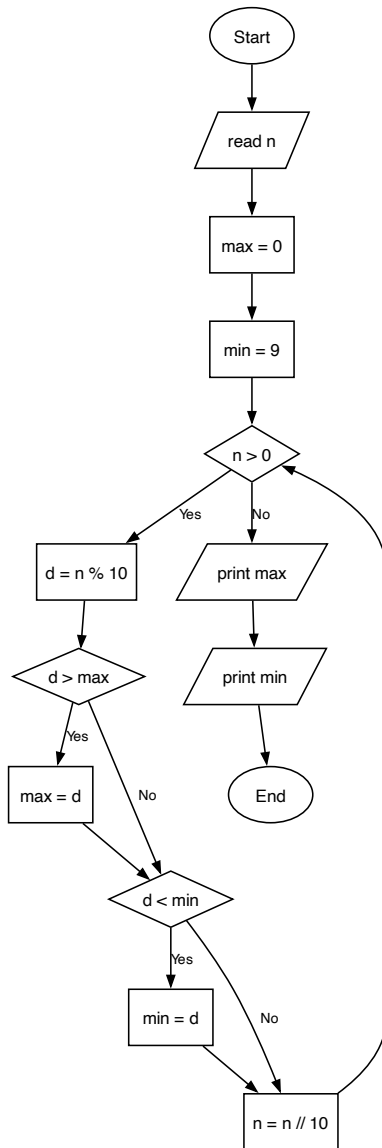


What picture appears? The whole image is just 8 bytes — 64 ones and zeros. Roughly how many bytes would a picture twice as wide and twice as tall (16 × 16) need?

The picture: a heart

Problem 8.

Trace this algorithm for $n = 5184$, and complete the trace table.



n	max	min	d	output
5184				
5184	0			
5184	0	9		
5184	0	9	4	
5184	4	9	4	
5184	4	4	4	
518	4	4	4	
518	4	4	8	
518	8	4	8	
51	8	4	8	
51	8	4	1	
51	8	1	1	
5	8	1	1	
5	8	1	5	
0	8	1	5	
0	8	1	5	8
0	8	1	5	1

Output: 8, 1